

Statistics for HCI

Alan Dix

alan@hcibook.com

Computational Foundry, Swansea University

Swansea, Wales, UK

Cardiff Metropolitan University

Cardiff, Wales, UK

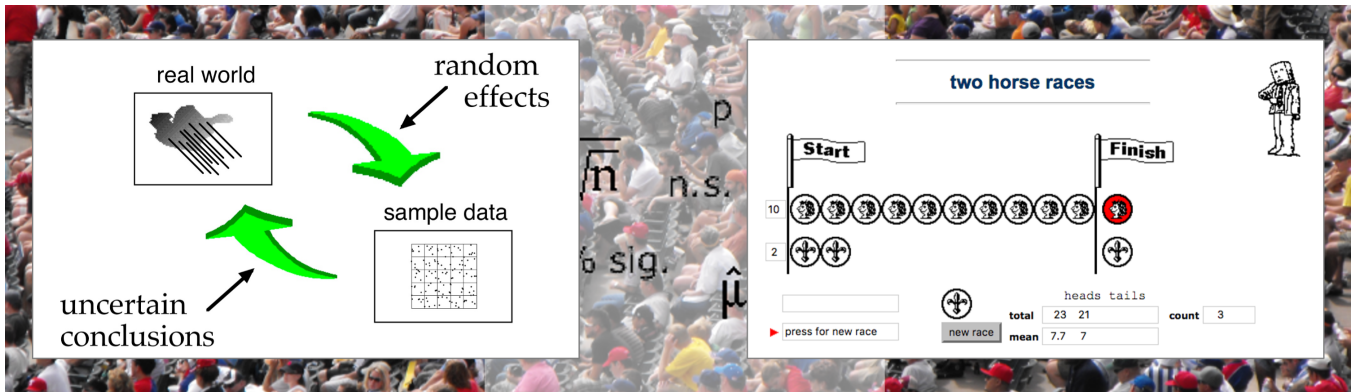


Figure 1: Sample materials: (left) the job of statistics; (right) interactive web experiments.

Background: Crowd on Flickr by razorray15, <https://www.flickr.com/photos/36714668@N02/4598065240>

ABSTRACT

Many researchers and practitioners find statistics confusing. This course aims to give attendees an understanding of the meaning of the various statistics they see in papers or need to use in their own work. The course builds on the instructor's previous tutorials and master classes including at CHI 2022, and on his recently book "Statistics for HCI: Making Sense of Quantitative Data". The course will focus especially on material you will not find in a conventional textbook or statistics course including aspects of statistical 'craft' skill, and offer attendees an introduction to some of the instructor's extensive online material.

CCS CONCEPTS

• **Human-centered computing** → HCI design and evaluation methods; • **Mathematics of computing** → Probability and statistics.

KEYWORDS

Statistics; human-computer interaction; evaluation; hypothesis testing; Bayesian statistics; statistical crisis

ACM Reference Format:

Alan Dix. 2023. Statistics for HCI. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems (CHI EA '23)*, April 23–28, 2023, Hamburg, Germany. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3544549.3574185>

1 BENEFITS

This course is intended to fill the gap between the 'how to' knowledge in basic statistics courses and having a real understanding of what those statistics mean. It will also help attendees to make sense of the various alternative approaches presented in recent articles in HCI and wider scientific literature (e.g. [1, 2, 5–7]).

At the end of the course attendees will have a richer understanding of: the nature of random phenomena and different kinds of uncertainty; the different options for analysing data and their strengths and weaknesses; ways to design studies and experiments to increase 'power' – the likelihood of successfully uncovering real effects; and the pitfalls to avoid and issues to consider when dealing with empirical data. Attendees will leave better able to design studies that efficiently use resources available and appropriately, effectively and reliably analyse the results

2 INTENDED AUDIENCE(S)

The course is intended for both experienced researchers and students who have already, or intend to engage in quantitative analysis of empirical data or other forms of statistical analysis. It will also be of value to practitioners using quantitative evaluation.

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).
CHI EA '23, April 23–28, 2023, Hamburg, Germany
© 2023 Copyright held by the owner/author(s).
ACM ISBN 978-1-4503-9422-2/23/04.
<https://doi.org/10.1145/3544549.3574185>

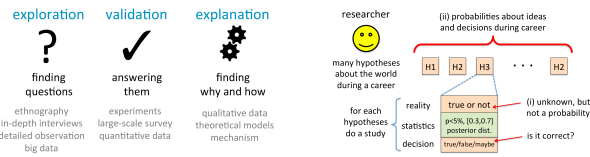


Figure 2: Example materials: (left) Research goals; (right) Probabilities of researcher's success or specific hypothesis.

3 PREREQUISITES

The course will assume some familiarity with statistical concepts theoretical or practical, for example, the use of t-tests or similar techniques. There will be occasional formulae, but the focus of the course is on conceptual understanding not mathematical skills.

4 CONTENT

The instructor has a wide range of videos and materials freely available online; so this course will follow the T-model [3] offering a broad road-map of the area, delving into depth in a few topics. If attendees leave feeling mastery of selected material and know what is available, they can fill in gaps in a 'just in time' manner when they wish.

4.1 Overview – key topics

The course as arranged around three key topics:

Understanding randomness – getting a **gut feel for the behaviour of random phenomenon** and in particular how easy it is to see structure when there is none – learning how probability can help us where our intuition fails – understanding the ubiquity of the Normal distribution and why it cannot be applied without great care to **power-law data from social networks** and similar phenomena.

Doing it (if not p then what?) – traditional (p-testing) and Bayesian statistics – what they mean and what they don't – philosophical differences between the two – common issues including the **role of the researcher vs the experiment**.

Design and implementation – gaining power – how to ensure studies and experiments reveal real effects including the **noise-effect-number triangle** – interpreting and visualising results.

We'll look at these main topics in detail below, but, following the T-model, the intention is not to dig into all equally. The highlighted aspects above are likely areas to drop into depth as they are least common in standard 'how to do it' treatments. However, I'd also like to offer opportunities for participants to ask questions, so will weight up presented content vs. discursive elements. The ultimate aim is not to cover loads of content – the online materials provide that anyway, but to give the participants the sense that this is something they can engage with themselves.

4.2 Getting started – why

The course will begin by asking first whether you need statistics anyway, and also understanding the *job of statistics* (Fig. 1, left), how it enables us to probe the curious world between probability and the utterly unknown.

Crucially, we will think about different stages in an investigation and consider the appropriate techniques for each. For iterative design do you need *formative* or *summative* evaluation? For research is your intention to test an existing hypothesis (*validation*) or to find out what you should be looking for (*exploration*)? For both there are times when you need an *explanation* in order to more deeply understand the problems you are working on (see Fig. 2, right).

4.3 Wild and wide

It is hard to get an instinctive feel for random things. So we will join in some exercises and demonstrations of the **unexpected wildness of random phenomena** including the effects of bias and non-independence (when one result effects others).

We will discuss different kinds of distribution and the reasons why the normal distribution (classic hat shape), on which so many statistical tests are based, is so common. In particular we will look at some of the ways in which the effects we see in HCI *may not satisfy* the assumptions behind the normal distribution.

Most will be aware of the use of non-parametric statistics for discrete data such as Likert scales, but there are other ways in which non-normal distributions arise. Positive feedback effects, which give rise to the beauty of a snowflake, also create effects such as the bi-modal distribution of student marks in certain kinds of university courses (don't believe those who say marks should be normally distributed!). This can become more complex if feedback processes include some form of threshold or other non-linear effect (e.g. when the rate of a task just gets too much for a user).

All of these effects are found in the processes that give rise to social networks both online and offline and other forms of network phenomena, which are often far better described by a long-tailed **power law**.

4.4 Doing it – if not p then what

There are several major kinds of statistical analysis methods:

- Hypothesis testing (the dreaded p!) – robust but confusing
- Confidence intervals – powerful but underused
- Bayesian stats – mathematically clean but fragile
- Empirical and simulation based – rare but useful

None of these is a magic bullet; all need care and a level of statistical understanding to apply.

We will discuss how these are related including the relationship between 'likelihood' in hypothesis testing and conditional probability as used in Bayesian analysis. There are common issues including the need to clearly report numbers and tests/distributions used, avoiding cherry picking, dealing with outliers, non-independent effects and correlated features. However, there are also specific issues for each method.

Classic statistical methods used in hypothesis testing and confidence intervals depend on ideas of 'worse' for measures, which are sometimes obvious, sometimes need thought (one vs. two tailed test), and sometimes outright confusing. In addition, care is needed in hypothesis testing to avoid classic fails such as treating non-significant as no-effect and inflated effect sizes. In Bayesian statistics different problems arise: the need to be able to decide in a robust and defensible manner what are the expected likelihood

of different hypothesis before an experiment; and the dangers of common causes leading to inflated probability estimates due to an initial fluke event. Crucially, while all methods have problems that need to be avoided, not using statistics at all can be far worse!

There are many common misunderstandings about both traditional and Bayesian statistics. One way to help makes sense of these is by focusing on the **role of the researcher** rather than the phenomenon being studied itself. Significance testing p-values, confidence interval percentages, and Bayesian posterior probabilities can be semantically incoherent if interpreted as probabilities on the phenomenon (which are typically simply true or not – unknown), but do make sense as probability statements about the accuracy of a researcher, laboratory or discipline over time (Fig. 2, right).

4.5 Design and implementation

A central problem in HCI studies is the dreaded 'too few participants'. To address this I offer a number of approaches towards **gaining power**.

Statistical power is about whether an experiment or study is likely to reveal an effect if it is present. Without a sufficiently 'powerful' study, you risk being in the middle ground of 'not proven', not being able to make a strong statement either for or against whatever effect, system, or theory you are testing. In HCI studies the greatest problem is often finding sufficient participants to do meaningful statistics. For professional practice we hear that 'five users are enough', but less often that this figure was based on particular historical contingencies and in the context of single *iterations*, not summative evaluations, which still need the equivalent of 'power' to be reliable.

However, power arises from the *noise-effect-number triangle* a combination of the size of the effect you are trying to detect, the size of the study (number of trials/participants) and the size of the 'noise' (the random or uncontrolled factors). Increasing the number of participants is not the only way to increase power and we will discuss various ways in which careful design, selection of subjects and tasks can increase the power of your study albeit sometimes requiring care in interpreting results. For example, using a very narrow user group can reduce individual differences in knowledge and skill (reduce noise) and make it easier to see the effect of a novel interaction technique, but also reduces generalisation beyond that group. In another example, we will also see how careful choice of a task can even be used to deal with infrequent expert slips.

Supposing you have completed your experiment or study and have your data – what next, **how do you make sense of the results**? In fact one of the best ways to design a study is to imagine this situation before you start! There are number of questions to think about during analysis (or design). Is it a one-off study, or part of a process (e.g. '5 users' for iterative development)? How to make sure your results and data can be used by others (e.g. repeatability, meta analysis)? Looking at the data, and asking if it makes sense given your assumptions (e.g. Fitts' Law experiments that assume index of difficulty without checking it holds in their own situation). Thinking about the conditions – what have you really shown – some general result or simply that one system or group of users is better than another?

5 PRACTICAL WORK

There are occasional practical exercises, e.g. coin tossing experiments ... but no complex numerical calculations!

6 INSTRUCTOR BACKGROUND

Alan Dix is Professor and Director of the Computational Foundry at Swansea University and is Professorial Fellow at Cardiff Metropolitan University, in both roles working across disciplines to encourage fundamental research that makes a difference to real people. He is well known for his HCI textbook and research in HCI including CSCW, mobile interfaces, technical creativity, and some of the earliest work on privacy and ethical implications of intelligent data processing. More recent work includes community engagement especially in rural areas and his one thousand mile research walk around Wales, which generated substantial quantitative and qualitative open research data from blogs to biodata.

However, before he was in HCI, Alan was a mathematician, including representing the UK in the International Mathematical Olympiad. He has practiced as a professional statistician and applied mathematician including work on modelling agricultural crop sprays, medical statistics and undersea cable detection. Within HCI these skills have been applied in his foundational work on formal methods for interactive systems, the use of Bayesian techniques in education, random sampling for visualisation of big data and uncertainty, and analysis of potential bias against human/applied areas in REF, the UK research assessment exercise.

This unusual combination of skills and experience gives Alan a unique insight into the issues and problems when applying statistics to HCI data, which he has recently embodied in a recent Morgan & Claypool volume "Statistics for HCI: making sense of quantitative data" [4].

7 RESOURCES

The tutorial will connect with "Statistics for HCI: Making Sense of Quantitative Data" [4], but will not require the attendees to have the book. There is also an extensive website of open materials including notes, slides, videos, interactive demos (e.g. Fig 1, right) and an online glossary: <http://alandix.com/statistics>

8 ACCESSIBILITY

Slides are detailed, to help those with reduced hearing or non-native English listeners. Some graphs are used, but explained verbally also. Online versions of coin tossing are available as an alternative for less dexterous participants. Extensive, videos, slides and notes are also available to allow attendees to recap after the course.

REFERENCES

- [1] Monya Baker. 2016. Statisticians issue warning over misuse of *P* values. *Nature News* 531, 7593 (March 2016), 151. <https://doi.org/10.1038/nature.2016.19503>
- [2] Andy Cockburn, Carl Gutwin, and Alan Dix. 2018. HARK No More: On the Preregistration of CHI Experiments. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems* (Montreal QC, Canada) (CHI '18). Association for Computing Machinery, New York, NY, USA, 1–12. <https://doi.org/10.1145/3173574.3173715>
- [3] Alan Dix. 2003. Mastery. *SIGCHI Bull.: Suppl. Interactions* 2003 (March 2003), 7. <https://doi.org/10.1145/967199.967209>
- [4] Alan Dix. 2020. *Statistics for HCI: Making Sense of Quantitative Data*. Morgan & Claypool. <https://doi.org/10.2200/S00974ED1V01Y201912HCI044>

- [5] John P. A. Ioannidis. 2005. Why Most Published Research Findings Are False. *PLoS Med* 2, 8 (08 2005), e124. <https://doi.org/10.1371/journal.pmed.0020124>
- [6] Maurits Kaptein and Judy Robertson. 2012. Rethinking Statistical Analysis Methods for CHI. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Austin, Texas, USA) (CHI '12). Association for Computing Machinery, New York, NY, USA, 1105–1114. <https://doi.org/10.1145/2207676.2208557>
- [7] Matthew Kay, Gregory L. Nelson, and Eric B. Hekler. 2016. Researcher-Centered Design of Statistics: Why Bayesian Statistics Better Fit the Culture and Incentives of HCI. In *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems* (San Jose, California, USA) (CHI '16). Association for Computing Machinery, New York, NY, USA, 4521–4532. <https://doi.org/10.1145/2858036.2858465>